Agenzia nazionale per le nuove tecnologie,
l'energia e lo sviluppo economico sostenibile

Mini Challenge

Irradiance prediction with ML

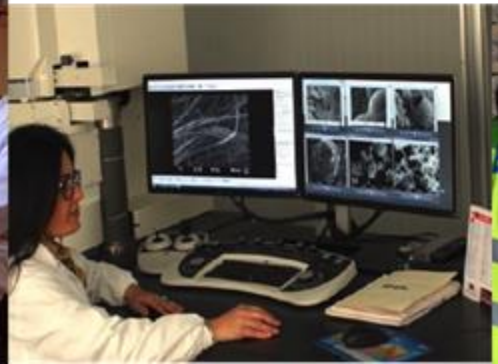
May 9th, 2025

Sofia Dutto - Gabriele Piantadosi










Agenzia Nazionale per le nuove tecnologie, l'energia e lo sviluppo economico sostenibile

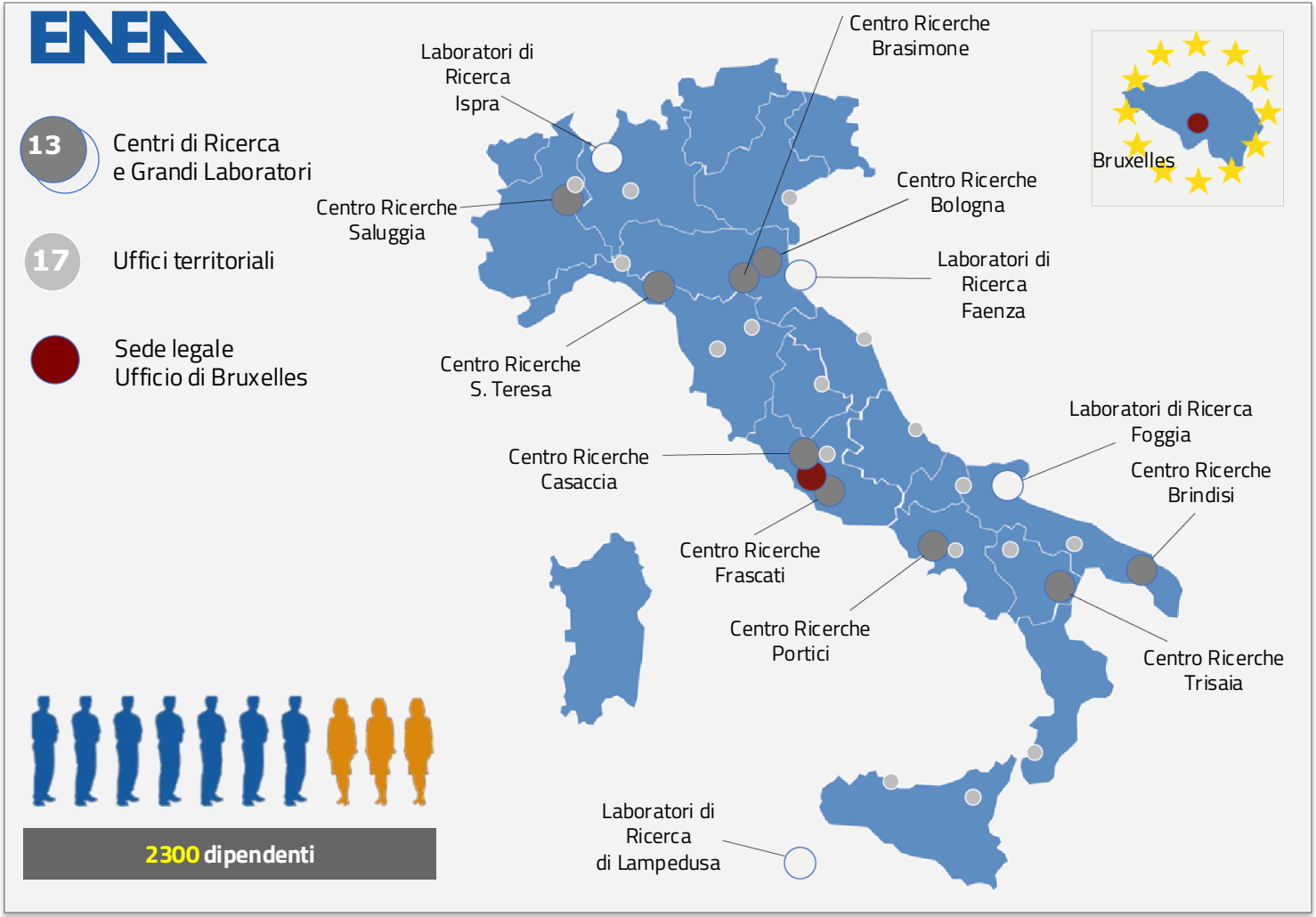
La nostra mission

"L'ENEA è l'Agenzia nazionale per le nuove tecnologie, l'energia e lo sviluppo economico sostenibile, ente di diritto pubblico finalizzato alla ricerca, all'innovazione tecnologica e alla prestazione di servizi avanzati alle imprese, alla pubblica amministrazione e ai cittadini nei settori dell'energia, dell'ambiente e dello sviluppo economico sostenibile (art. 4 Legge 28 dicembre 2015, n. 221)."

Sin dalla nascita negli anni '60, i nostri punti di forza sono la ricerca applicata, il trasferimento tecnologico e l'assistenza tecnico-scientifica a imprese, associazioni, territori, amministrazioni centrali e locali.






13
Centri di Ricerca
e Grandi Laboratori
17
Uffici territoriali
Sede legale
Ufficio di Bruxelles
Laboratori di Ricerca Ispra
Centro Ricerche Brasimone
Bruxelles
Centro Ricerche Saluggia
Centro Ricerche Bologna
Laboratori di Ricerca Faenza
Centro Ricerche S. Teresa
Laboratori di Ricerca Foggia
Centro Ricerche Casaccia
Centro Ricerche Brindisi
Centro Ricerche Frascati
Centro Ricerche Portici
Centro Ricerche Trisaia
Laboratori di Ricerca di Lampedusa
2300 dipendenti

Le nostre sfide



Contribuire alla transizione energetica ed ecologica attraverso lo sviluppo tecnologico, dell'innovazione, di servizi avanzati per promuovere la crescita e la competitività in settori quali:

- Economia circolare
- Cambiamento climatico
- Protezione dell'ambiente, della salute, dei beni culturali
- Energia, dalla produzione all'efficienza negli usi finali

Come siamo organizzati

PRESIDENTE
Gilberto Dialuce

Consiglio di amministrazione

Direttore Generale

4 UNITA' - 1 ISTITUTO

- Relazioni e Comunicazione
- Istituto di Radioprotezione
- Studi, Analisi e Valutazioni
- Missioni in Antartide
- Organi di Vertice

4 DIPARTIMENTI

- Fusione e Sicurezza Nucleare
- Tecnologie Energetiche e fonti rinnovabili
- Sostenibilità dei Sistemi Produttivi e Territoriali
- Efficienza Energetica

6 DIREZIONI

- Board dei Direttori
- Amministrazione Centrale
- Personale
- Affari legali e Trasparenza
- Infrastrutture e Servizi
- Innovazione, sviluppo e trasferimento tecnologico

Dipartimento tecnologie energetiche e fonti rinnovabili

MISSION contribuire allo sviluppo di un mix energetico climate-friendly, di tecnologie innovative per la decarbonizzazione, la mobilità sostenibile, l'idrogeno green, le fuel cells, il solare termico e termodinamico, il fotovoltaico, i biocombustibili, le biomasse, le smart grids, le supergrids, le smart cities e le smart communities

ENEA è il coordinatore del Cluster Nazionale Energia

AREE DI ECCELLENZA

- **Fonti rinnovabili**
- **Idrogeno Green e fuel cell**
- **Mobilità sostenibile**
- **Smart grids, Super grids**
- **Smart cities**
- **Comunità energetiche**





Infrastrutture di ricerca

- **ZECOMIX Zero Emission of Carbon with MIXed technologies**, impianto pilota nel Centro di ricerca ENEA Casaccia
- **PIBE, Piattaforma per la bioenergia , la bioraffineria e la chimica verde** Centro di ricerca ENEA Trisaia
- **CRESCO Supercomputer**
  High Performance Computing
  Centro di ricerca ENEA Portici
- **CSP Plant - Concentrated Solar Power**
  Impianto dimostrativo per il solare a concentrazione
  Centro di ricerca ENEA Casaccia

GRANDI PROGETTI

- **AB-Lab, Advanced Battery Laboratory,** Laboratorio per lo sviluppo delle batterie – Investimento da 27 milioni di euro Centro di Ricerca ENEA Casaccia
- **ENEA Hydrogen Valley**, Polo per la sperimentazione di una filiera nazionale dell'idrogeno. Investimento da 14 milioni di euro Centro di ricerca ENEA Casaccia



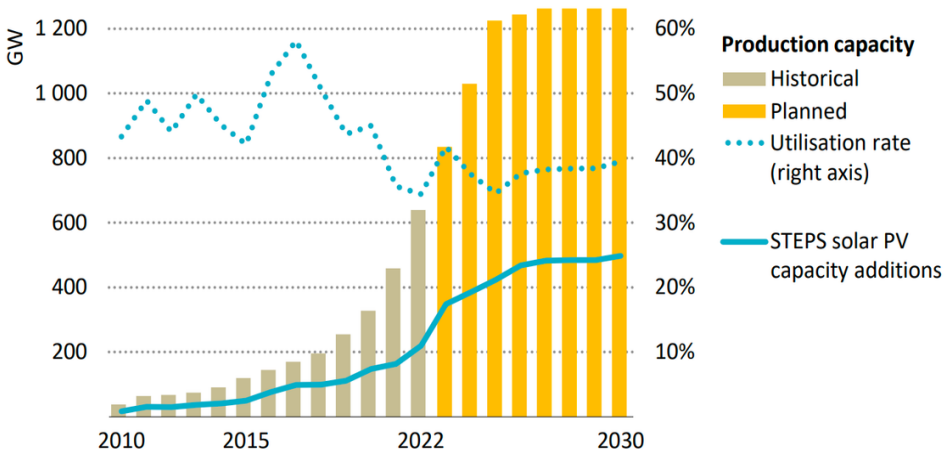
Introduction

The use of solar energy has seen a notable expansion in the last ten years. This trend is expected to continue at a high pace, with investments planned to increase global PV module production capacity **from around 640 GW in 2022 to over 1,200 GW** in the medium term (IEA, 2023).

The possibility to fully exploit the production capacity of PV plants represents a huge opportunity to promote the use of solar energy worldwide and to accelerate the clean energy transition.

However, the **uncertainty** on PV energy production and the difficult integration with the main power grid slow down PV market development.

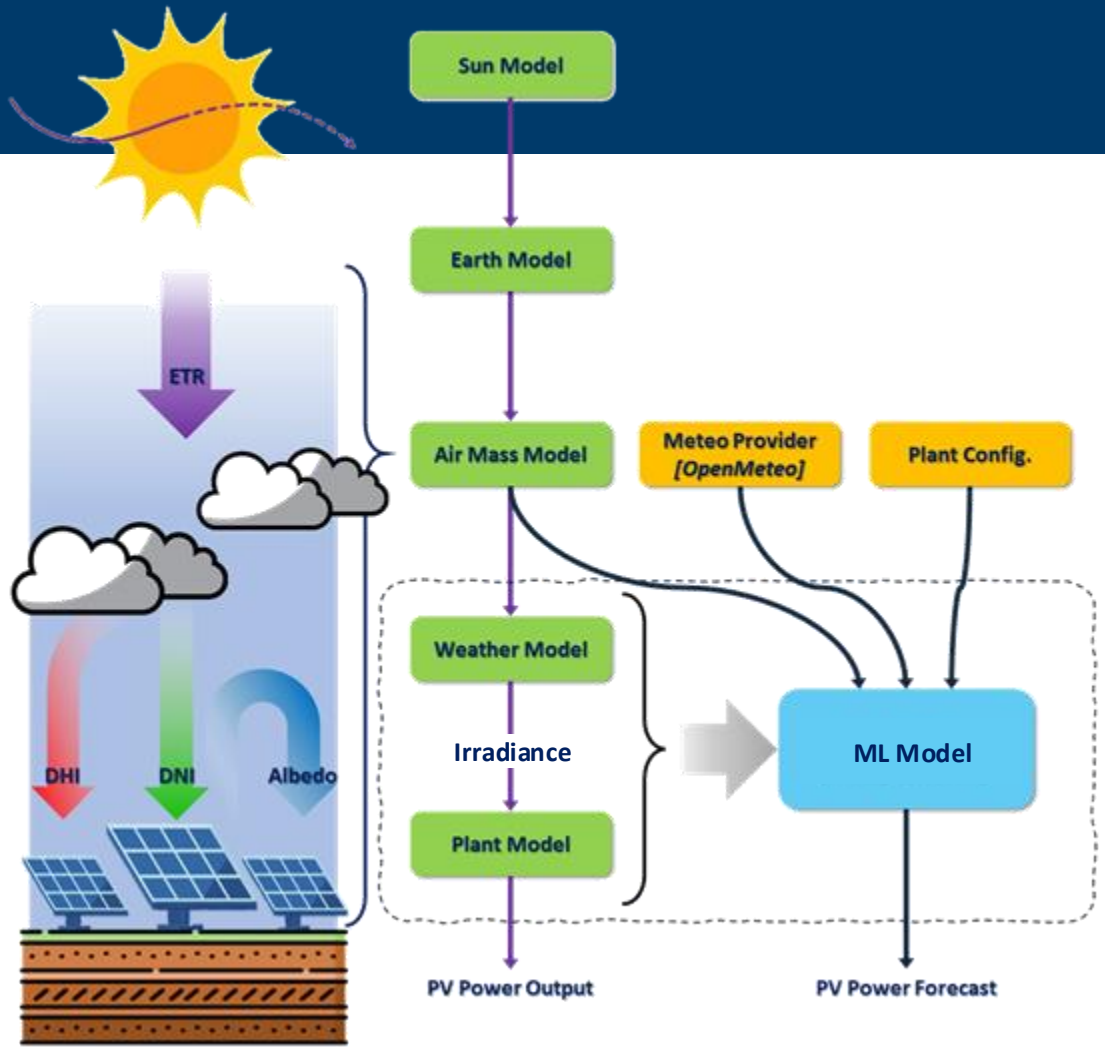
Global solar module manufacturing and solar PV capacity additions in the STEPS, 2010-2030



IEA. CC BY 4.0.

source: World Energy Outlook 2023 (International Energy Agency)

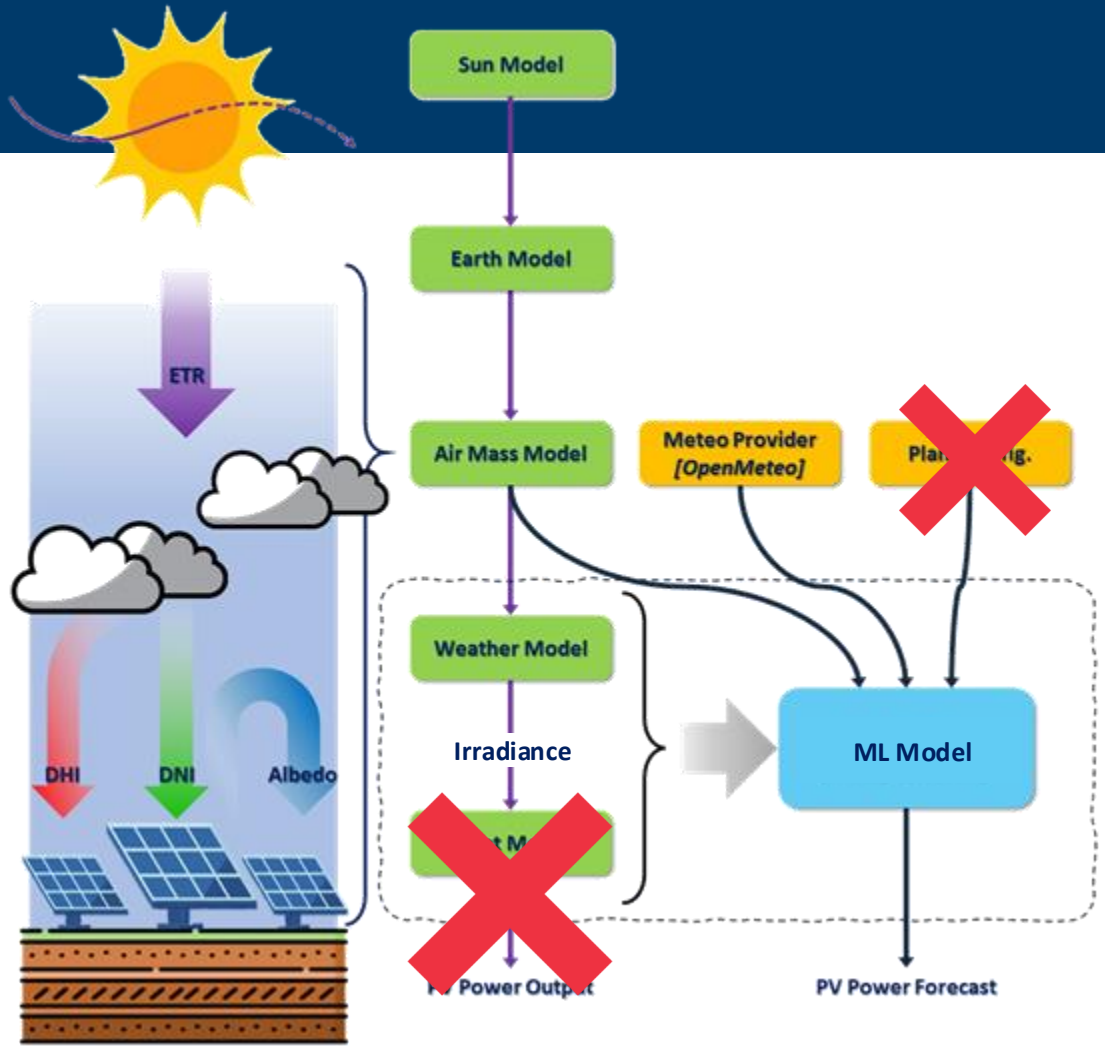
Our work



Accurate PV production prediction requires understanding the **phenomena and processes affecting yield,** from solar radiation generation to electricity conversion, including extraterrestrial radiation, air mass modeling, and weather conditions.

Most predictive models use **locally measured irradiance** as a feature to bypass the complexity of modeling such phenomena, simplifying the process. Relying on partial observations or approximations can result in a cascade of errors, potentially degrading the performance of predictive models over time.

Your work

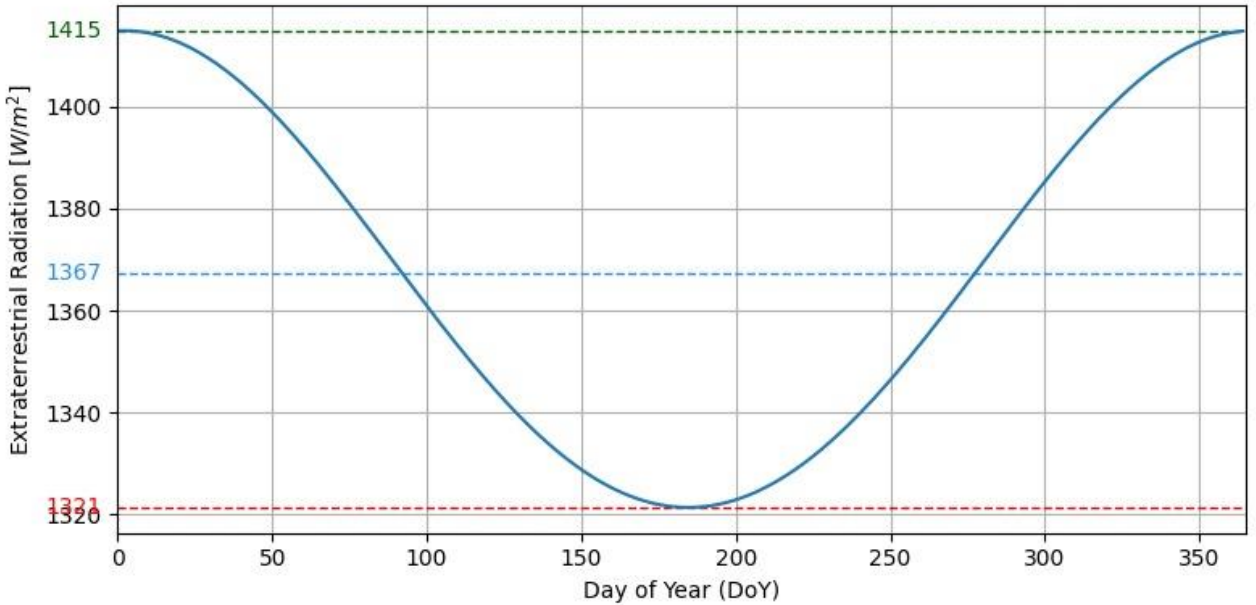


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What is irradiance?

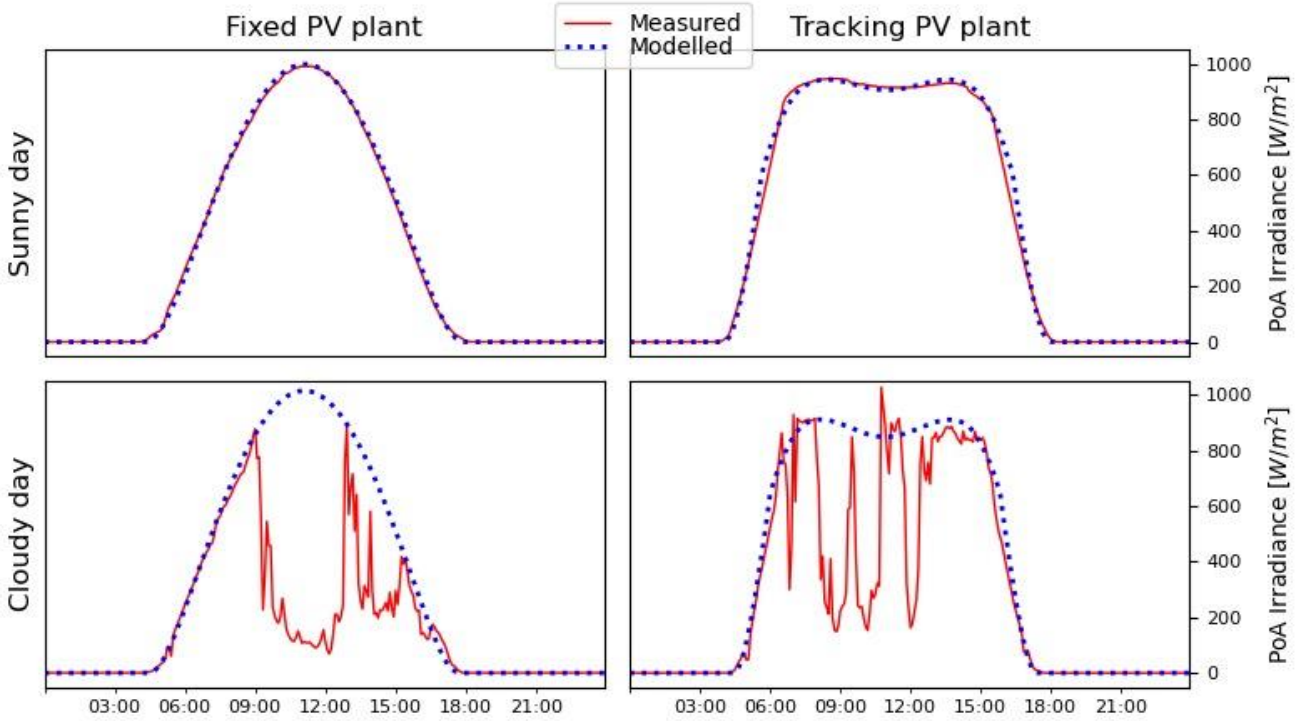
Extraterrestrial radiation over the course of the year, with an average value of approximately 1367 W/m². The actual solar irradiance received by a photovoltaic panel is typically lower due to atmospheric attenuation, panel orientation and local meteorological conditions.



Irradiance by PV-lib – Clear Sky Irradiance Model

pvlib

https://pvlib-python.readthedocs.io/en/stable/user_guide/clearsky.html



Weather Variables



OpenMeteo

https://open-meteo.com/en/docs/historical-weather-api

Hourly Weather Variables

- [x] Temperature (2 m)
- [x] Relative Humidity (2 m)
- [x] Dewpoint (2 m)
- [x] Apparent Temperature
- [x] Precipitation (rain + snow)
- [x] Rain
- [x] Snowfall
- [ ] Snow depth
- [ ] Weather code
- [x] Sealevel Pressure
- [x] Surface Pressure
- [x] Cloud cover Total
- [x] Cloud cover Low
- [x] Cloud cover Mid
- [x] Cloud cover High
- [x] Reference Evapotranspiration (ET_0)
- [x] Vapour Pressure Deficit
- [x] Wind Speed (10 m)
- [x] Wind Speed (100 m)
- [x] Wind Direction (10 m)
- [x] Wind Direction (100 m)
- [x] Wind Gusts (10 m)
- [x] Soil Temperature (0-7 cm)
- [x] Soil Temperature (7-28 cm)
- [x] Soil Temperature (28-100 cm)
- [x] Soil Temperature (100-255 cm)
- [x] Soil Moisture (0-7 cm)
- [x] Soil Moisture (7-28 cm)
- [x] Soil Moisture (28-100 cm)
- [x] Soil Moisture (100-255 cm)

Data Overview (OpenMeteo weather)

Variable	Unit	Description
temperature_2m	°C	Air temperature at 2 meters above ground
relative_humidity_2m	%	Relative humidity at 2 meters above ground
dew_point_2m	°C	Dew point temperature at 2 meters above ground
apparent_temperature	°C	Apparent temperature is the perceived feels-like temperature combining wind chill factor, relative humidity and solar irradiance
precipitation	mm	Total precipitation (rain, showers, snow) as sum of the preceding hour.
rain	mm	Only liquid precipitation of the preceding hour including local showers and rain from large scale systems.
snowfall	cm	Snowfall amount of the preceding hour in cm.
pressure_msl surface_pressure	hPa	Atmospheric air pressure reduced to mean sea level or pressure at surface.
cloud_cover	%	Total cloud cover as an area fraction
cloud_cover_low	%	Low level clouds and fog up to 2 km altitude
cloud_cover_mid	%	Mid level clouds from 2 to 6 km altitude
cloud_cover_high	%	High level clouds from 6 km altitude
et0_fao_evapotranspiration	mm	ET_0 Reference Evapotranspiration of a well watered grass field. ET_0 is calculated from temperature, wind speed, humidity and solar irradiance.

Data Overview (OpenMeteo weather)

vapour_pressure_deficit	kPa	Vapor Pressure Deificit (VPD) in kilopascal (kPa). For high VPD (>1.6), water transpiration of plants increases. For low VPD (<0.4), transpiration decreases
wind_speed_10m wind_speed_100m	km/h km/h	Wind speed at 10 or 100 meters above ground. Wind speed on 10 meters is the standard level.
wind_direction_10m wind_direction_100m	°	Wind direction at 10 or 100 meters above ground
wind_gusts_10m	km/h	Gusts at 10 meters above ground of the indicated hour. Wind gusts in CERRA are defined as the maximum wind gusts of the preceding hour.
soil_temperature_0_to_7cm soil_temperature_7_to_28cm soil_temperature_28_to_100cm soil_temperature_100_to_255cm	°C	Average temperature of different soil levels below ground.
soil_moisture_0_to_7cm soil_moisture_7_to_28cm soil_moisture_28_to_100cm soil_moisture_100_to_255cm	m^3/m^3	Average soil water content as volumetric mixing ratio at 0-7, 7-28, 28-100 and 100-255 cm depths.
is_day	/	0 = night; 1 = day

Data Overview (PV-lib modelled irradiance)

Intermediate Variables

apparenth_zenith	°	Apparent sun zenith accounting for atmospheric refraction.
zenith	°	Solar zenith angle
apparent_elevation	°	Apparent sun elevation accounting for atmospheric refraction.
elevation	°	Actual elevation (not accounting for refraction) of the sun in decimal degrees, 0 = on horizon. The complement of the zenith angle.
azimuth	°	Azimuth of the sun in decimal degrees East of North. This is the complement of the apparent zenith angle.
equation_of_time	min	Difference in time between solar time and mean solar time.
ghi	W/m^2	Global horizontal irradiance
dni	W/m^2	Direct normal irradiance
dhi	W/m^2	Diffuse horizontal irradiance
tracker_theta	°	Tracker rotation angle as a right-handed rotation around the axis
aoi	°	Angle of incidence of the solar vector on a surface. This is the angle between the solar vector and the surface normal
surface_azimuth	°	Azimuth angle of the surface
surface_tilt	°	Panel tilt from horizontal. For example, a surface facing up = 0°, surface facing horizon = 90°
POAb	W/m^2	Component of direct sunlight that is incident on the surface of a PV module
POAg	W/m^2	Component of direct sunlight that is reflected off the ground and then strikes on the surface of a PV module
POAd	W/m^2	Component of direct sunlight that strikes on the surface of a PV module after having been scattered by molecules or particulates in the atmosphere.
Irr0	W/m^2	Modelled irradiance

Irr0 = POAb + POAb + POAb

Data Overview (measured irradiance - target)

Irr (target)	W/m^2	Total irradiance received on the PV plane of array as average of the preceding hour.

GOOD LUCK!

Any questions or doubts?

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